Introduction to Data Analytics
MS - 4610

Session 2 – Probability Review

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Overview of Probability Theory

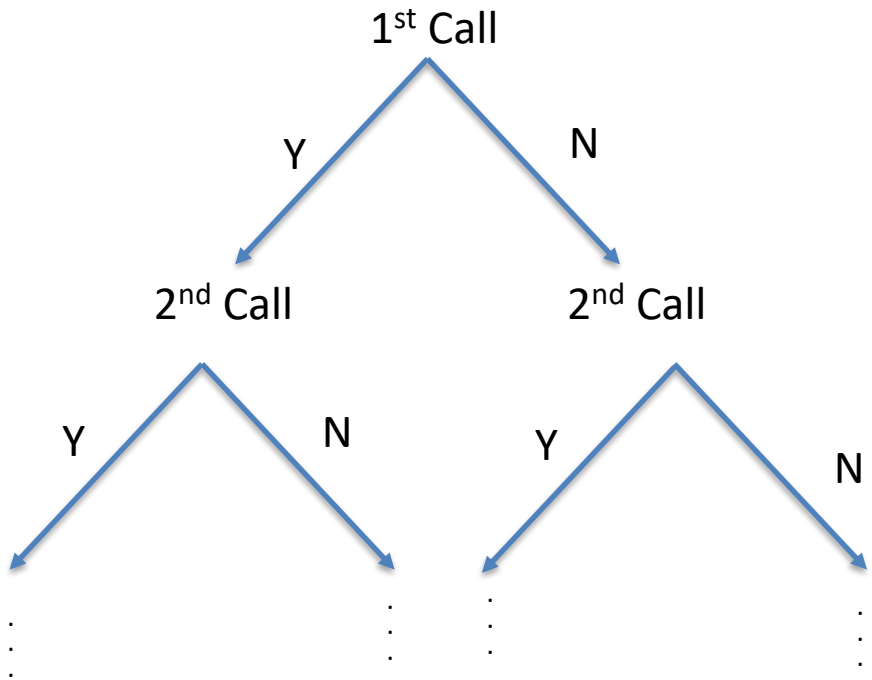
- Overview of this section
  - Introduction to Probability
    - Sample spaces and Events
    - Random Variables
    - Distributions – Discrete and Continuous
    - Probability Mass/Density function
    - Cumulative Distribution Functions
    - Mean and Variance of Probability distributions
    - Functions of random variables and joint distributions
  - Common Distributions
    - Bernoulli, Uniform, Binomial, Poisson, Geometric, Exponential, Normal

Sample Spaces and Events

- Random Experiment: An experiment that can result in different outcomes, even though it is repeated in the same manner every time.
- Sample spaces: The set of all possible outcomes of a random experiment is called a sample space.
  - Discrete and continuous spaces

Sample Spaces and Events

- Example: Making 5 sales calls



{YYYYY, YYYYN, YYYNY, YYYNN, …. NNNNN}

Sample Space and Events

- What if we only cared about the number of successful sales.
- Then the state space becomes {0,1,2,3,4,5}
- Event: An event is a subset of the sample space of a random experiment
- Probability of an event is the likelihood of choosing the subset

Sample Space and Events

- Using set operation on Events: Union ($A \cup B$), Intersection ($\mathrm{A} \cap B$), and Complement (A')
- Example: Union of getting 4 sales and 5 sales: {YYYYN, YYYNY..} U {YYYYY} answers the question of getting at least 4 sales
- When two events share nothing in common (have a null intersection) they are said to be mutually exclusive

Random Variables

- We defined *sample spaces*, and *events* with respect to random experiments.
- A random variable is characterized by a mathematical function that quantifies the likelihood (absolute or relative) of the outcomes in the sample space of a random experiment.
  - In the discrete case, the PMF reflects the probability that a specific value or a countable set of values can occur.
  - In the continuous case, the PDF reflects probability of occurrence within a given range or interval.
  - All distributions can be characterized by CDF which captures the probability of occurrences below a certain value x.
- Notation: X denotes the Random Variable, and x denotes the outcome (variates)

Discrete and Continuous random variables

- Examples of discrete: Number of visitors/customers, Number of defects/ defective parts, Number of complaints received or accidents, grades of anything (this course or credit rating).
- Examples of continuous: Time taken, length/width/area/volume, profit in Rupees or as ROI, proportion of a certain chemical.
- Having too many discrete states, and the discretization of a continuous random variable.

Discrete Random Variables

- Probability Mass function (PMF): For a discrete random variable X with possible values $x_1, x_2, \dots, x_n$, a probability mass function is such that: $f(x_i) = P(X = x_i)$ and

a) $0 \leq f(x_i) \leq 1$, b) $\sum_{i=1}^{n} f(x_i) = 1$

Example: $\mathrm{f}(x) = \frac{2x+1}{25}, x = 0,1,2,3,4$

- Cumulative distribution function is denoted by: $F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$ and

a) $0 \leq F(x) \leq 1$, b) if $x \leq y$ then $F(x) \leq F(y)$

Discrete Random Variables

- Mean: A measure of the central tendency

$$\mu = E(X) = \sum_{x} x f(x)$$

- Variance: A measure of the dispersion

$$\sigma^{2} = E(x-\mu)^{2} = \sum_{x} (x-\mu)^{2} f(x)$$

Which can be written as $\sum_{x} x^{2} f(x) - \mu^{2}$

The standard deviation of x is $\sigma = \sqrt{\sigma^{2}}$

Continuous Random Variables

- Probability Density function (PDF):
- $P(a \leq x \leq b) = \int_a^b f(x).dx$ = area under f(x) from a to b; and

a) $f(x) \geq 0$, b) $\int_{-\infty}^{\infty} f(x) = 1$, c) The probability at a given point is 0

- Cumulative distribution functions

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(u).du$$

Continuous Random Variables

- Mean: A measure of the central tendency of the random variable

$$E(x) = \mu = \int_{-\infty}^{\infty} x f(x) dx$$

- Variance: A measure of the dispersion

$$\text{Var}(x) = \sigma^2 = \int_{-\infty}^{\infty} (x-\mu)^2 f(x).dx$$

or $\int_{-\infty}^{\infty} (x)^2 f(x).dx - \mu^2$

Axioms - continued

- For two events, A and B, $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(\mathrm{B}) - P(A \cap B)$ (addition rule)
- Conditional probability
  - $P(B|A) = \frac{P(A \cap B)}{P(A)}$
- Independence. Two events are independent if $P(A \cap B) = P(A) \,.\, P(B)$ or $P(B|A) = \mathrm{P}(\mathrm{B})$ or $P(A|B) = P(A)$
- Multiplication rule: $P(A \cap B) = P(B|A)P(A) = P(A|B)P(B)$ or Bayes theorem: $P(B|A) = \frac{P(A|B)P(B)}{P(A)}$

Functions of Random Variables

- Sometimes we are interested in a function of the random variable
  - Example1: If the number of people who visit our store is a random variable, our revenue is a function of this random variable.
  - Example 2: If the price of oil is random variable, then the price of petrol is a function dependent on this.
- The function of a random variable is another random variable which we can obtain from the original PMF or PDF

Functions of Random Variables

- Let $f_X(x)$ be the RV, and we are interested in the RV $Y = g(X)$
- Expected value can be straight forward $E(Y) = \int_{-\infty}^{\infty} g(x).f_X(x).dx$
- Steps for PDF:
  - Find CDF as $F_Y(y) = P(Y \leq y) = P(g(X) \leq y) = P(X \leq g^{-1}(y))$
  - If CDF is continuous then $f_Y(y) = \frac{d}{dy}(F_Y(y))$
- Example: Let X be uniform on [0,1]. Find PDF of $Y = \sqrt{X}$

$$F_Y(y) = P(Y \leq y) = P\left(\sqrt{X} \leq y\right) = P(X \leq y^2) = y^2$$

Therefore $f_Y(y) = 2y, 0 \leq y \leq 1$

Conditioning on Random Variables

- We saw that $P(B|A) = \frac{P(A \cap B)}{P(A)}$
- You can use that to compute (numerically) the probability of B occurring given that A occurs. What if you wanted a conditional PDF?
- $f_{X|A}(x) = \begin{cases} \frac{f_X(x)}{P(X \in A)} \text{ if } x \in A, \\ \text{and 0 otherwise} \end{cases}$

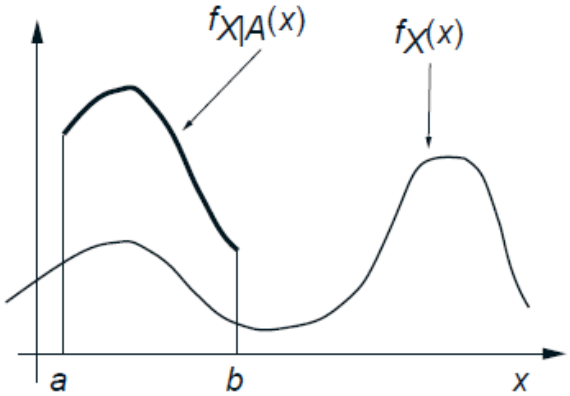


Fig 3.13 from Bertsekas, Dimitri P., and John N. Tsitsiklis. *Introduction to probability*. Vol. 1. Belmont, MA: Athena Scientific, 2002.

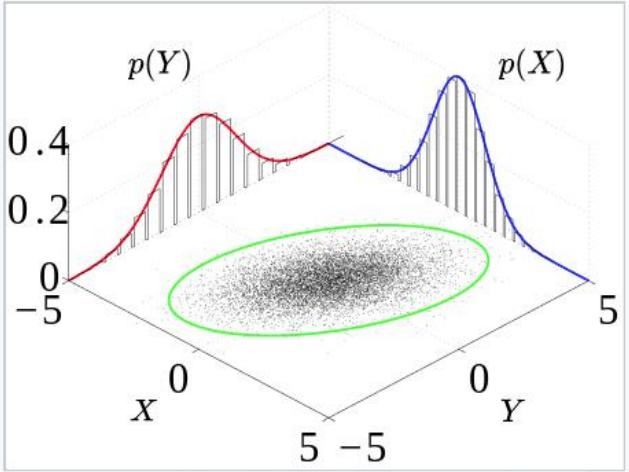
Joint random variables

- Why not just keep them separate? The photocopy shop example.
- Two random variables are jointly continuous/discrete and can be described by a joint PDF or PMF $f_{X,Y}(x, y)$
- PDF $f_{X,Y}(x, y)$ captures $P(a \leq X \leq b, c \leq Y \leq d) = \int_c^d \int_a^b f(x, y).dx.dy$
- PMF: $f_{X,Y}(x, y)$ captures $P(X = x \; and \; Y = y)$

Continuous case

- How do I imagine this:

- $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y).dx.dy = 1$
- $f_X(x) = \int_{-\infty}^{\infty} f(x,y).dy$ (marginals)
- $f_{X|Y}(x|y) = \frac{f(x,y)}{f(y)}$ (conditionals)



Many sample observations (black) are shown from a joint probability distribution. The marginal densities are shown as well.

Wikipedia contributors. "Joint probability distribution." *Wikipedia, The Free Encyclopedia*. Wikipedia, The Free Encyclopedia, 10 Jul. 2018. Web. 1 Aug. 2018.